build-a-bot

take back your tools!!

Peckham Digital 2026

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workshop plan

- Intro – 5 mins
- Background – 15 mins
- Ideation – 10 mins
- Making (+ more ideas?!) – 45 mins
- Round up – 15 mins

Look at what bots can be, design one, and build it on your own laptop.

before we start

templates and guide: mahaliahr.github.io/diy-bot

open **Ollama**, then `check.html`, and follow it

Open terminal:

```
ollama pull ministral-3:3b
```

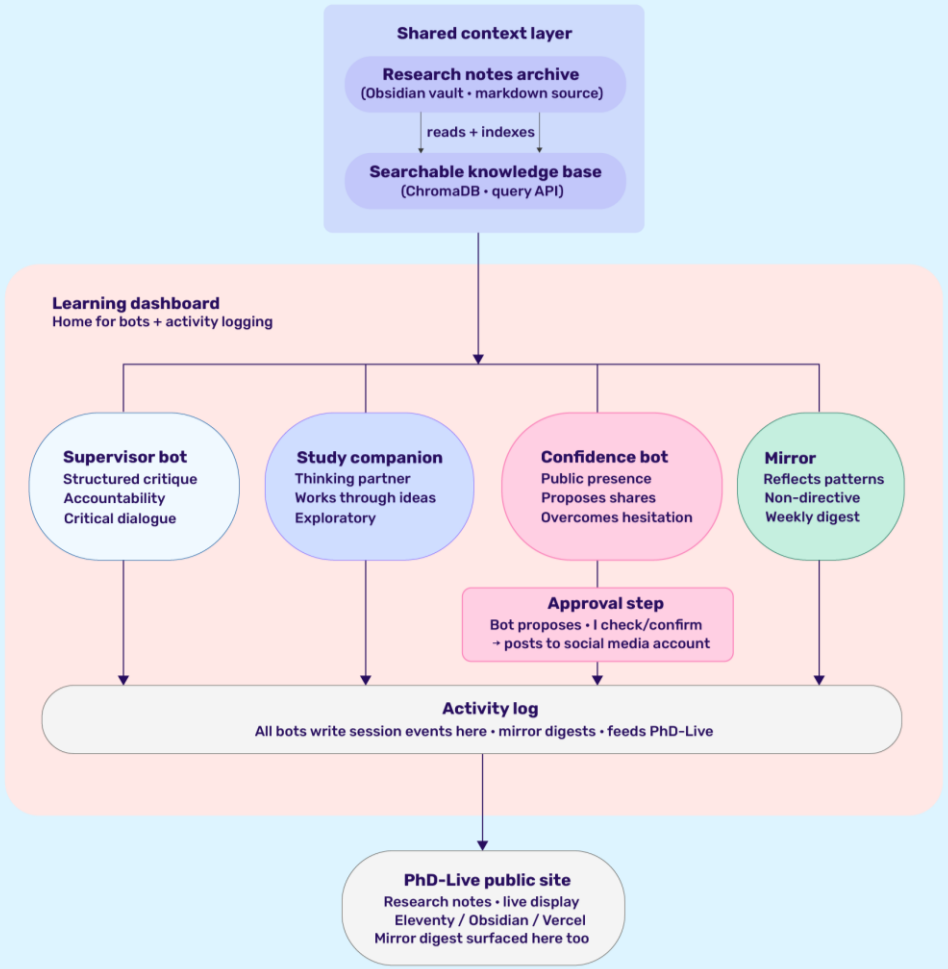
about me

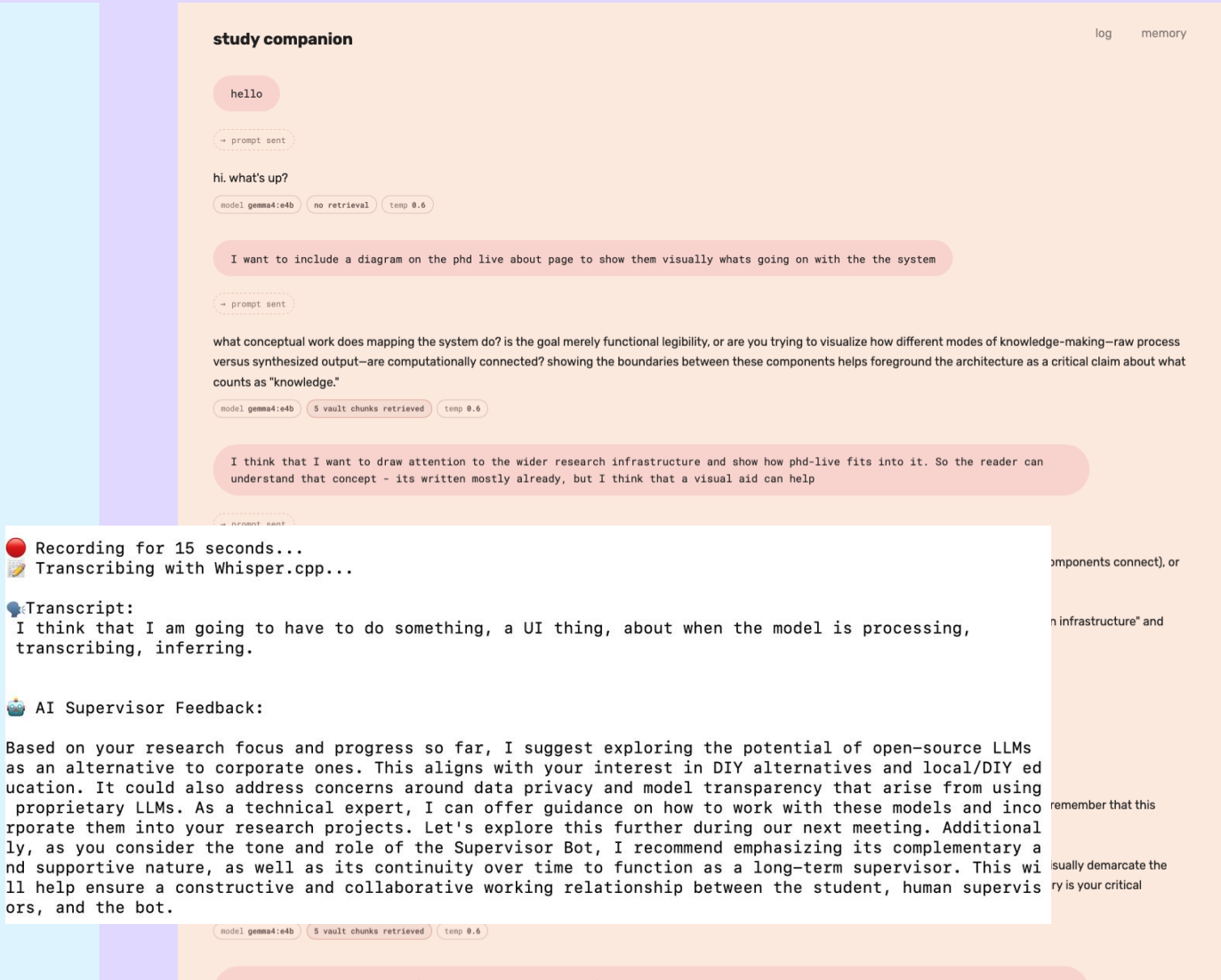
artist, designer, live coder

Senior Lecturer, CCI, UAL

PhD: ’Building Possibility Spaces for Creative Learning, With and Through Generative Systems’

What are the potentials of a local LLM system in creative learning?





what we are building today

Giving you the building blocks of how to create your bots that you can connect to your own systems

this is the beginning of a journey…

You can go really deep, but the start is accessible to anyone

background

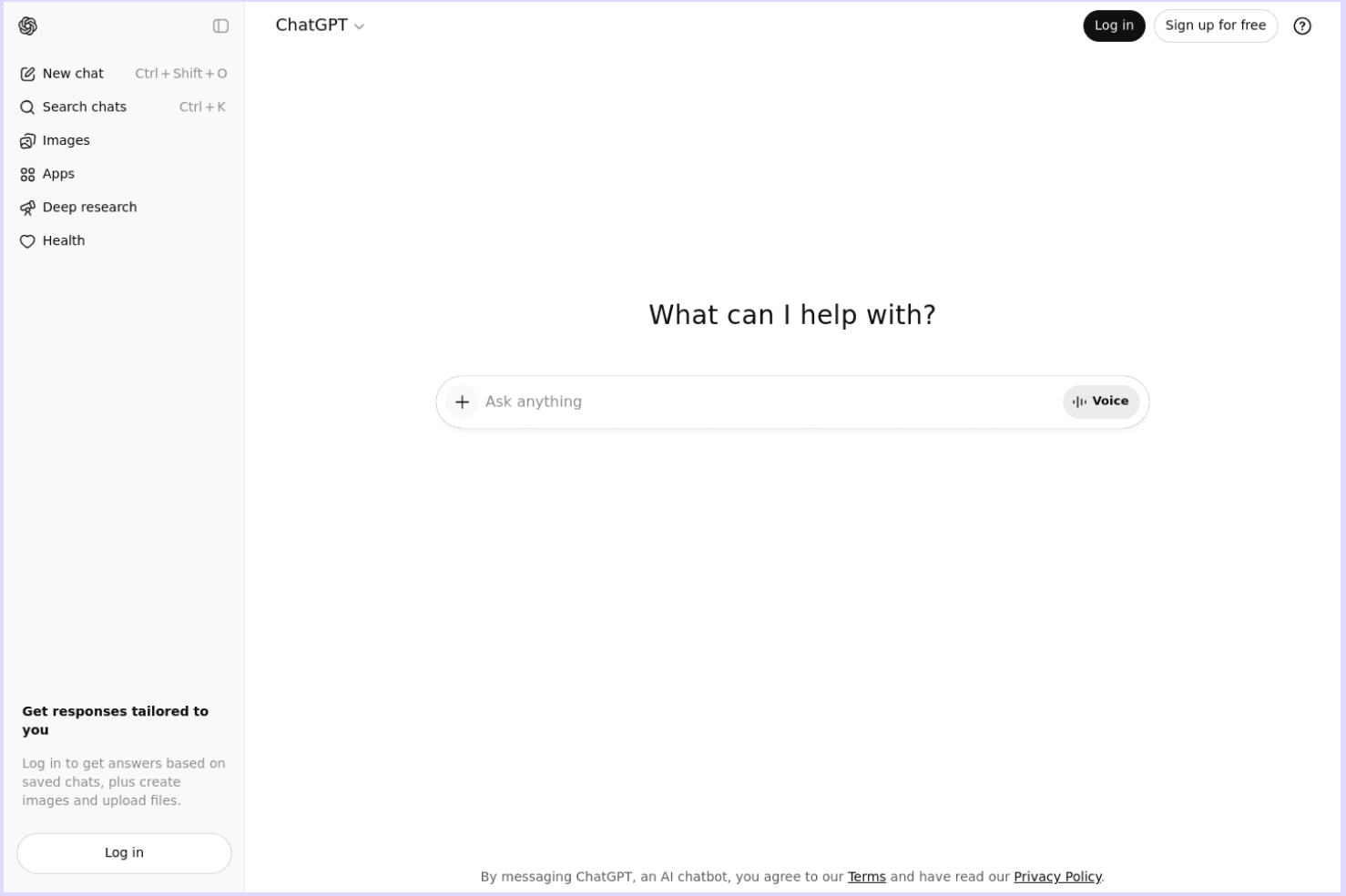
15 minutes

what is a bot?

what is a bot?

something that acts automatically, on someone's behalf

the public imagination



why do people use chatbots?

(question for the room)

performing knowledge

It doesn't know.

It predicts what an answer would sound like.

taking a chatbot apart

before you ask done once, by the company

collect

text gathered from the web, books and forums, often without asking

→

train

a model learns the patterns in it: a large file of numbers

→

shape

more training, so it acts like a helpful assistant: polite, confident

each time you ask

hidden instructions

written by the company, added to every conversation

+

your message

and the conversation so far

+

sometimes: found text

search results or documents, added before it answers

→

predict

the trained model writes the most likely reply, word by word

dashed = you can't see it.
in the templates, nothing in the second row is hidden: open "prompt sent to model".

taking a chatbot apart



1 **interface**

what you see and touch: the page (chat.html, style.css)

2 **prompt**

what shapes its behaviour: your instructions and the conversation, shown in full

3 **model**

what writes the replies: a file on this laptop, picked from the menu

the same three layers are in every chatbot. here, none of them are hidden.

local versus cloud

cloud: your words go to a company's computers

local: the model lives on your laptop

what does open source mean here?

	download and run it	see what it learned from	use and change it freely
fully open Olmo, Apertus	yes	yes	yes
open weights Mistral, Gemma, Qwen	yes	no	mostly
restricted Llama	yes	no	on Meta's terms
closed ChatGPT, Claude, Gemini	no	no	no: only through their app

open isn't yes or no. it's what you're allowed to see, and to do.

my bots

Study Companion: shows the prompt it sends

Mirror: reflects, never writes to my notes. Or suggests anything new.

Supervisor Bot: listens, prompted to act as an advisor

other kinds of bots

web crawlers

the most common bots online

how training text was gathered, without asking

robots.txt: a polite request, not a lock

other kinds of bots



grow_slow

Nicole He

a webcam posts a photo of a plant every day

engine: a schedule. no AI

nicole.pizza/grow-slow



alias

Bjørn Karmann, Tore Knudsen, 2018

sits on a smart speaker and decides when it can listen

engine: a small model, on the device

bjoernkarmann.dk/project/project_alias



not the only one (N'TOO)

Stephanie Dinkins, around 2018

an AI built from three generations of her family's stories

engine: a model trained with care

stephaniedinkins.com

bot does not (necessarily) equal AI

a bot can run on rules, a schedule, or an LLM model

the model is one possible engine

a bot that acts for you

rules you can see
sources you choose
running where you can see it
trade-off: smaller, vaguer, breaks rules

questions

anything so far?

ideation

10 minutes

ideation

https://github.com/mahaliahr/diy-bot/archive/refs/heads/main.zip

your bot blueprint

- name and personality
- purpose
- what it reads, and never reads
- how you interact with it
- what it never does
- who owns the data
- where it runs

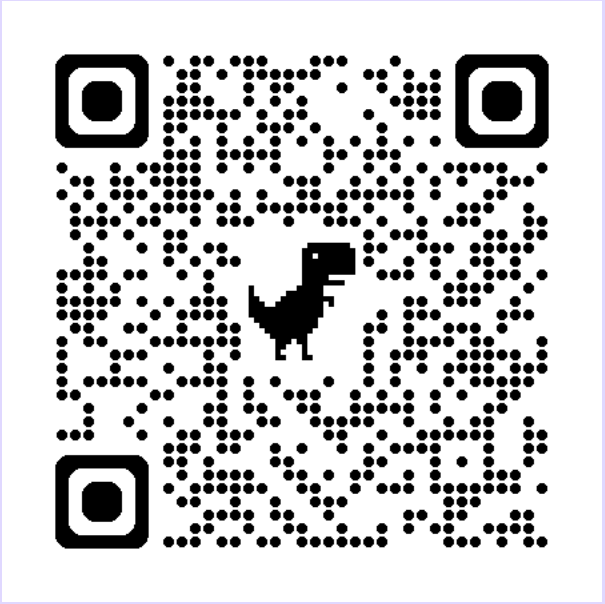
ways a bot can behave

behaviour		as an instruction
gives one provocation, never an answer	→	"Reply with one sentence that questions an assumption in what I wrote."
only ever asks questions	→	"Reply with one short question and nothing else."
only rearranges what you give it	→	"Use only words from my message. Add no new words."
replies like a postcard	→	"Start with 'Dear you,'. Under 50 words. End with a made-up place."

what would you call each one?

making

45 minutes



https://github.com/mahaliahr/diy-bot

what's in the folder

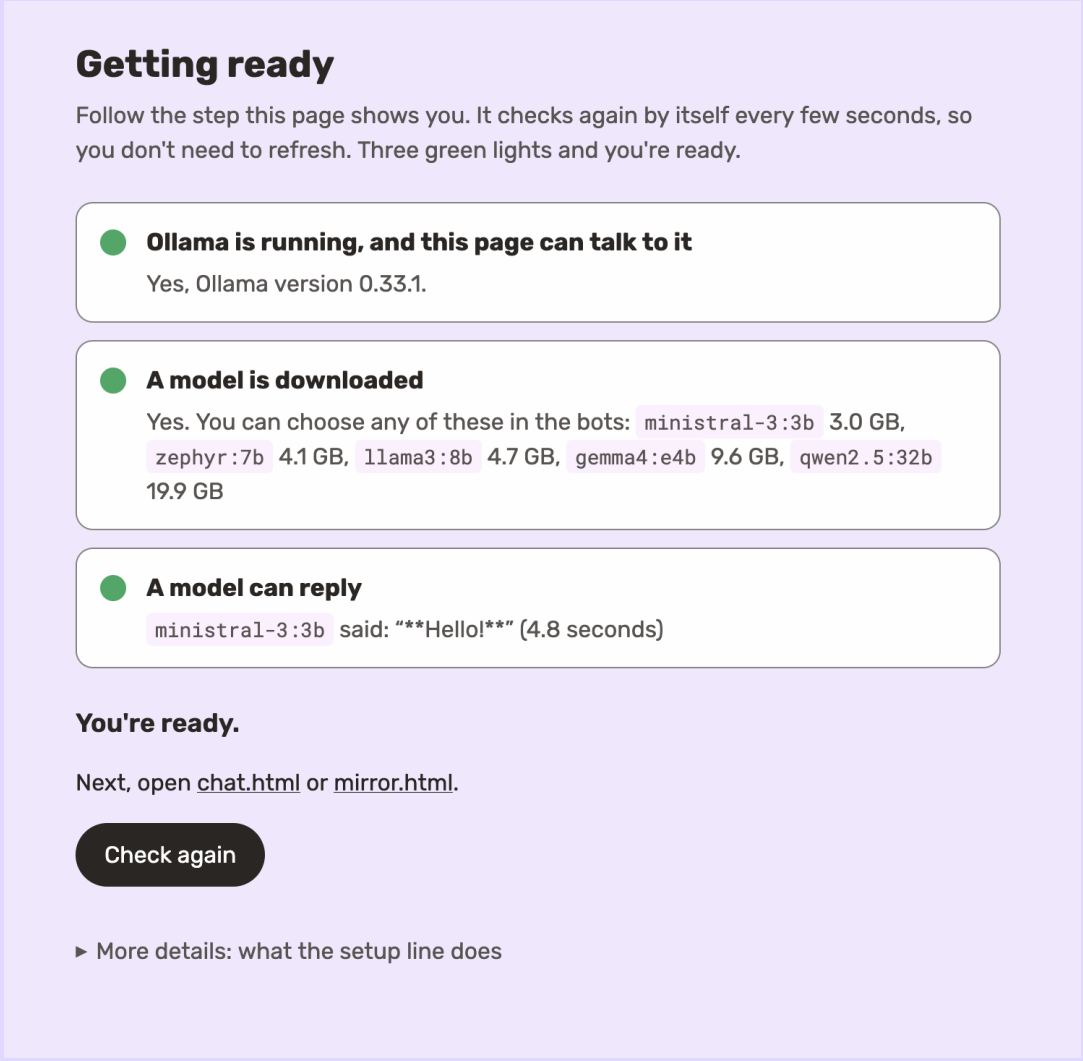
interface: `chat.html`, `mirror.html`, `style.css`

prompt: the **YOUR BOT** section

model: the menu on the page

check.html: is everything ready?

step 1: three green lights



open `check.html`

a red light says what to do

step 2: open the file in a text editor

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>DIY Bot: chat</title>
<link rel="stylesheet" href="style.css">
</head>
<body class="chat">

<div class="typo" id="typo" role="alert" hidden></div>
<script>
// If the code below has a typo, show a message instead of a blank page.
window.addEventListener("error", (event) => {
  if (!/syntax|unexpected|unterminated|missing|invalid|is not defined/i.test(event.message
  const box = document.getElementById("typo");
  box.hidden = false;
  box.textContent =
    "The bot can't start because of a typo in this file. Most often it's a missing backtic
    "or a missing quote or comma in the YOUR BOT section. Undo your last change, or look r
    event.lineno + ". Then save and refresh.";
});
</script>

<aside>
  <h1 id="bot-name">Bot</h1>

  <div class="setting">
```

Windows: Notepad

Mac: TextEdit, set to show HTML as code or VS Code

step 3: paste your blueprint

```
/* ================================================================
   YOUR BOT: this is the only part you need to change.
   Paste your blueprint into "instructions", save the file,
   then refresh this page in your browser.
   ================================================================ */

const BOT = {
  name: "Dot",                  // what your bot is called
  model: "ministral-3:mo3b",     // the model to start with. You can switch to any installed
  temperature: 0.7,             // 0 = predictable, 1 = surprising
  instructions: `You are Dot, a patient companion for creative work.
Your purpose is to help me think when I am stuck.
You only ever reply with one short question.
You never give advice, answers or lists.`
};
```

find **YOUR BOT**

replace the instructions

save, then refresh the page

instructions that work

1. one line: who it is and what it's for
2. most important rule first
3. say what to do, not only what not to
4. set the shape: length, tone, questions
5. unusual behaviour? show one example

for example

vague

Don't be long-winded.

better

Reply in two sentences, then ask one question.

Vague`

Be encouraging.

better

Start by naming one thing that's working.

small models follow short, plain instructions best.

step 4: talk to it and notice

Notice

open "prompt sent to model": what did it actually get?

where does it break its rules?

which rule is doing the most work?

test it like an experiment

keep three test messages:

- an ordinary one
- an unusual one
- one that tries to break a rule

change one thing at a time

still failing? try another model before rewording again

a rule a small model won't keep is a finding, not a failure.

step 5: same bot, different model

pick another model from the menu

what changes?

if your laptop struggles

pair up

share one laptop and build together

finding more models

ollama.com/library: search, then ollama pull

Hugging Face: many more, look for GGUF

check the size, 'instruct', and the model card

going further

colours and fonts: style.css

mirror: give it your own writing

`guide.html`: more ways to change it

share and reflect

What surprised you about how little it takes?

What was harder or easier than you expected?

take it home

templates and guide: mahaliahr.github.io/diy-bot

guide, section 3: after the workshop

Contact me if you want ☺ @mm_hr_